

Project Proposal: RideShare – A Ride Hailing System

1. Project Details

RideShare – Intelligent Ride Hailing Simulation System

Amail Uddin - 24K2575 – BDS3A

Hashir Vohra – 24K2531 – BDS3A

2. Project Overview

RideShare is a console-based ride-hailing simulation system inspired by platforms like Uber and Yango. The system demonstrates the integration of data structures, algorithms, and object-oriented programming concepts to simulate real-world ride-hailing operations including user management, route optimization, and dynamic fare calculation. Implemented in C++, it serves as an academic DSA semester project.

3. Problem Statement

In modern urban transportation, efficient ride allocation and route optimization are critical. However, simulating such systems for educational purposes is challenging due to their algorithmic complexity. This project aims to design a simplified yet realistic simulation of a ride-hailing platform that uses efficient data structures and algorithms to manage users, rides, and routes.

4. Objectives

- Design a structured and object-oriented simulation of a ride-hailing ecosystem.
- Apply graph algorithms for route optimization.
- Utilize hash maps, priority queues, and vectors for efficient data management.
- Demonstrate file I/O operations for persistent data storage.
- Analyze algorithmic efficiency and optimize time and space complexity.

5. System Features

1. User Management – Registration and login for both riders and drivers.
2. Driver-Rider Matching – Finds nearest available driver using a greedy + Dijkstra-based algorithm.
3. Shortest Path Calculation – Implements Dijkstra's/A* Algorithm for optimal route selection..
4. Ride Lifecycle Management – Handles all stages from request to completion.
5. Rating System – Riders rate drivers with updated averages.
6. Data Persistence – Saves and loads user and ride data using text files.

6. Conclusion

The RideShare system serves as a practical demonstration of how data structures and algorithms can model complex, real-world applications. It successfully integrates graph

theory, searching algorithms, and object-oriented design, making it an ideal semester project for a DSA course.